Lesson 4.5 — Simplifying Rational Expressions

Big idea: Simplifying a rational expression is just factoring the top and bottom and cancelling common factors. The cancelled factors leave behind domain restrictions.

Quick review

Factor everything first. You can only cancel FACTORS, not terms.

Cancel matching factors from numerator and denominator.

State excluded x-values from the original denominator. These restrictions persist even after simplification.

Watch for negatives: $(a - b) = -(b - a)$. Factor out the negative when needed.

Worked example

Worked example. Simplify $(x^2 - 9)/(x^2 - 5x + 6)$.

Factor: $(x - 3)(x + 3)/[(x - 2)(x - 3)] = (x + 3)/(x - 2)$, with $x \neq 3$ and $x \neq 2$.

Warm-ups

1. Simplify $(3x^2)/(6x)$.
2. Simplify $(x - 1)(x + 4) / (x - 1)$.
3. Simplify $(x^2 - 4)/(x + 2)$.

Practice

1. Simplify $(x^2 + 5x + 6)/(x^2 + x - 6)$.
2. Simplify $(2 - x)/(x - 2)$.
3. Simplify $(x^3 - 8)/(x^2 - 4)$. (Hint: difference of cubes on top.)

Challenge

1. Simplify $(x^3 + x^2 - 6x)/(x^2 - 4)$ and state restrictions.
2. Why is $(x - 2)/(2 - x)$ equal to -1 for all $x \neq 2$?

Answer key — Lesson 4.5

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $x/2$ ($x \neq 0$).
2. $x + 4$ ($x \neq 1$).
3. $(x - 2)(x + 2)/(x + 2) = x - 2$ ($x \neq -2$).

Practice

1. $(x + 2)(x + 3) / [(x + 3)(x - 2)] = (x + 2)/(x - 2)$, $x \neq \pm\ldots \Rightarrow x \neq 2, -3$.
2. $(2 - x) = -(x - 2) \Rightarrow -1$ ($x \neq 2$).
3. $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$; $x^2 - 4 = (x - 2)(x + 2)$. So $(x^2 + 2x + 4)/(x + 2)$, $x \neq \pm 2$.

Challenge

1. $x(x^2 + x - 6)/[(x - 2)(x + 2)] = x(x + 3)(x - 2)/[(x - 2)(x + 2)] = x(x + 3)/(x + 2)$, $x \neq \pm 2$.
2. Because $2 - x = -(x - 2)$, the fraction is $-(x - 2)/(x - 2) = -1$ (provided $x \neq 2$ so we're not dividing by zero).